ALICE L WALTON SCHOOL OF MEDICINE

Vancomycin and Bacitracin

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Adapted from: Cephalosporins, Vancomycin, and Bacitracin

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Learning Objectives (2) Versions

After completing this brick, you will be able to:

1. Explain the bactericidal mechanism of action of vancomycin and bacitracin.
2. Recognize the most common the adverse drug reactions associated with the use of vancomycin and bacitracin.

CASE CONNECTION

AY is a 55-year-old man in the Emergency room. He is complaining of warm redness on his right arm. About four days back, he had accidentally hit the wall outside his office while rushing to go home. After about 24 hours, he developed pain and started to see the redness in the area. He also noticed that his skin seemed 'torn' or broken' in that area. He said he had developed fever and chills. AY appeared ill, and his right arm had developed edema and was red (diffused) around the area of impact. There was an edematous appearance on his right arm. There was no crepitation and no purulence. The leg was very tender, and a skin break was noted. Elevated WBC counts of >16,000/mm^3, 15% band forms were noted. The two sets of blood cultures were drawn following hospital admission. They tested positive for group A β-hemolytic streptococci and MRSA.

How will you explain your treatment plan to AY? Consider your answer as you read, and we'll revisit DW at the end of the brick.

GO TO CONCLUSION ↓

What Are Vancomycin and Bacitracin?

Vancomycin and bacitracin are two popular peptidoglycan-inhibitor antibiotics, which just means that they help kill bacteria by breaking down their cell walls. Vancomycin is a glycopeptide often used to treat very serious staphylococcal infections, particularly those caused by methicillin-resistant *Staphylococcus aureus* (MRSA), while bacitracin is typically used as an ointment for superficial skin infections.

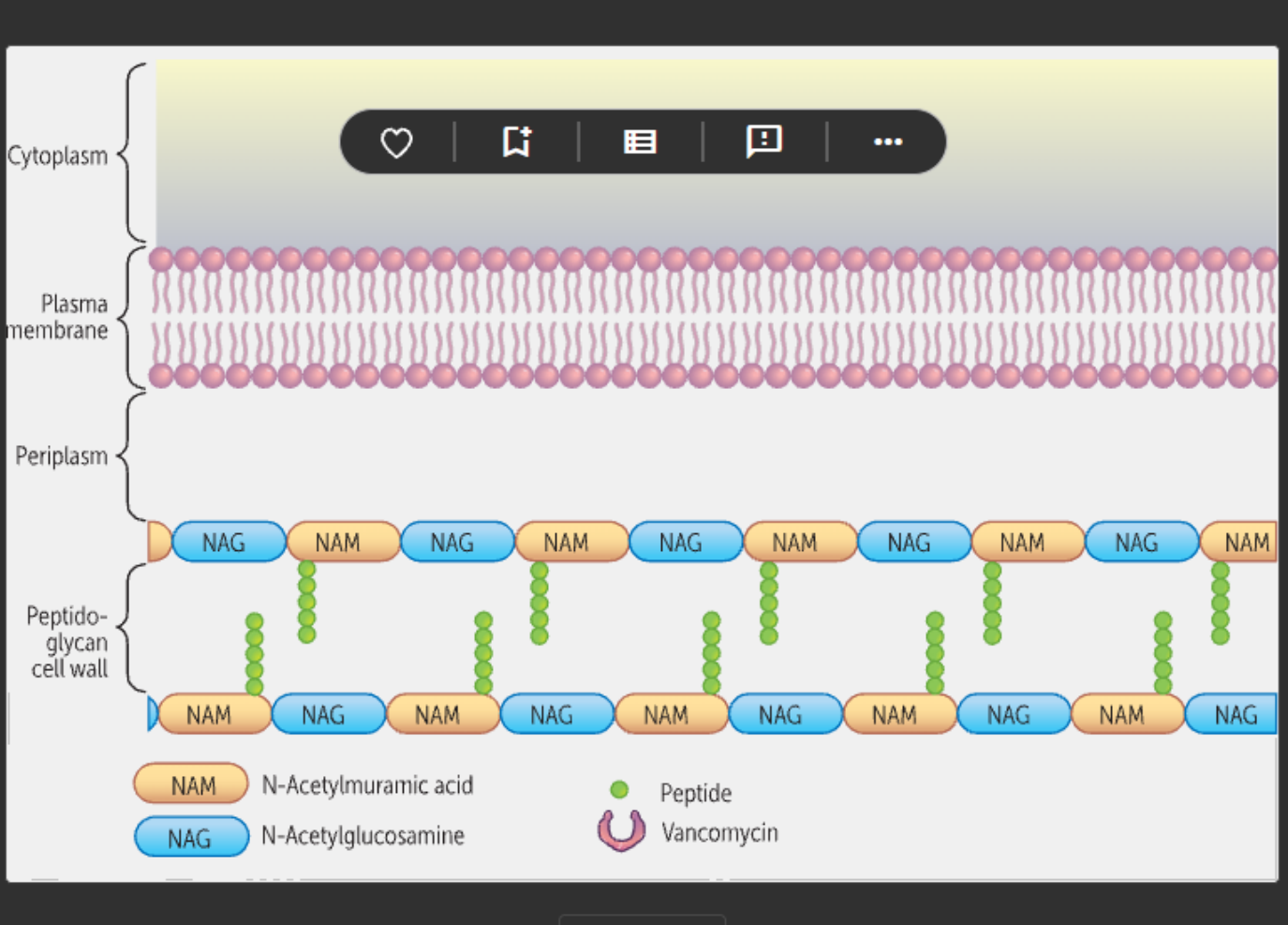
How do peptidoglycan inhibitors work?

How Do Vancomycin and Bacitracin Work? (LO#1)

These drugs all kill bacteria by altering the cell wall, but the ways in which they accomplish this alteration vary slightly. To better understand how these drugs work, let's first delve a little further into the details of bacterial cell wall synthesis.

Bacterial Cell Wall

The bacterial cell wall is made up of a substance called peptidoglycan, which can be divided into two components: a glycan, either N-acetyl-D-glucosamine (NAG) or N-acetylmuramic acid (NAM), and a peptide bridge (Figure 1).



QUIZ

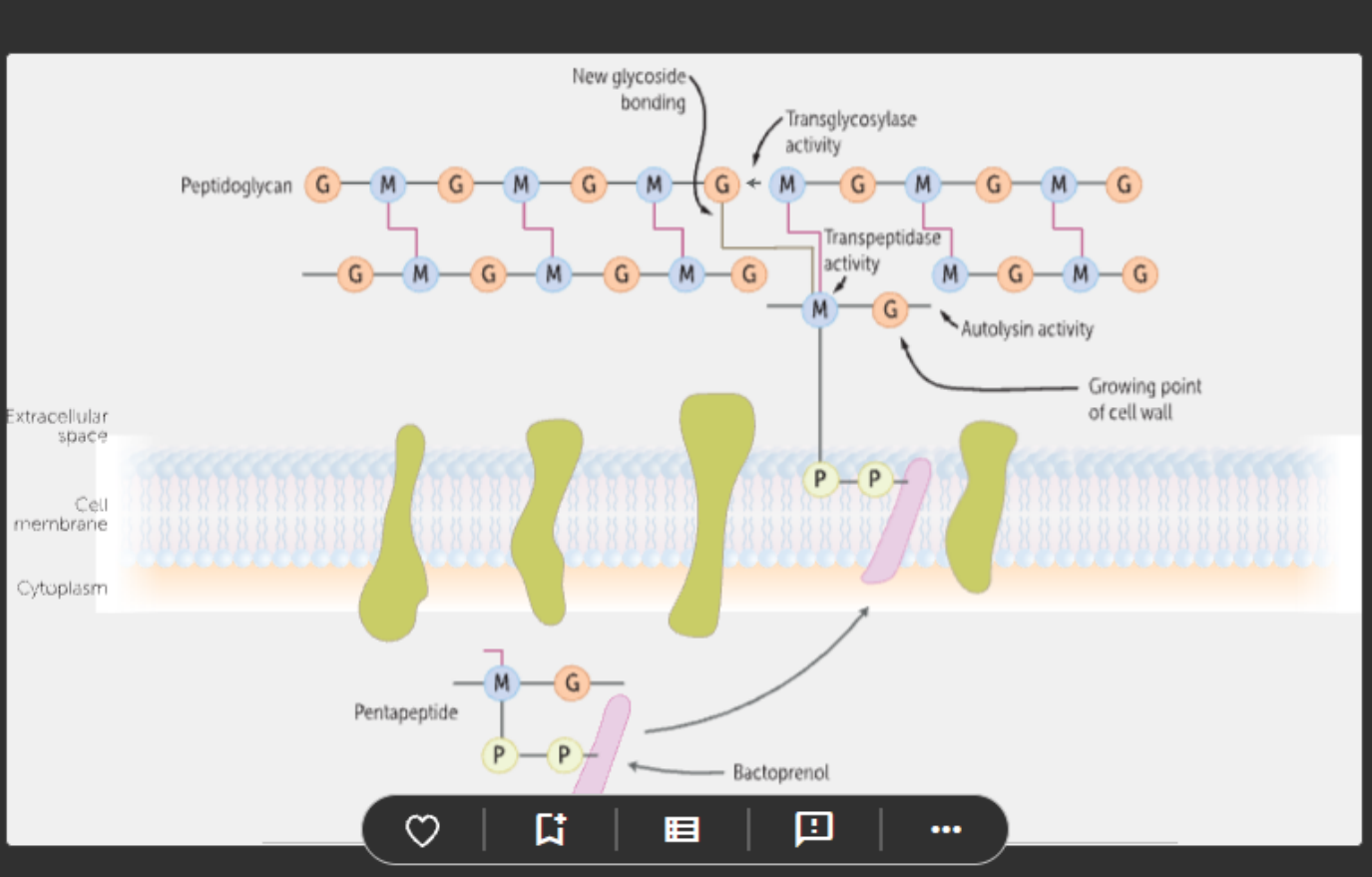
Figure 1

Credit: ©ScholarRx

What are the two components of peptidoglycan?

The glycans NAG and NAM are made in the cytoplasm of the cell and then eventually transported to the cell wall in a series of three steps:

1. First, NAM is joined to the lipid bactoprenol at the membrane surface (Figure 2).
2. NAG is then joined to the bactoprenol-NAM group.
3. The bactoprenol-NAM-NAG unit is transported across the membrane to the outer cell wall, where it acts as one additional unit to the growing peptidoglycan chain. Think of these as beads being added to a string.



QUIZ

Figure 2

Once these chains of peptidoglycan are constructed, they are cross-linked to each other through peptide chains by enzymes called transpeptidases. This allows multiple layers to develop in the cell wall, forming a rigid network.

Peptidoglycan Synthesis Inhibition

Vancomycin and bacitracin both inhibit peptidoglycan synthesis, but in different ways. Bacitracin works by preventing NAM from joining bactoprenol, as outlined above in step 1. This stops synthesis of any new units altogether! Vancomycin, on the other hand, exerts its effect a little further down the line—it stops the newly formed NAM–NAG unit from adding to the existing peptidoglycan chain, as mentioned in step 3. Vancomycin binds the D-Alanine (D-Ala D-Ala) terminus of the amino acid peptide, inhibiting cross-linkage of the cell wall. Vancomycin binds to the D-alanyl-D-alanine precursor and blocks peptidoglycan polymerization. The loss of this ability to add new beads to the string means that the peptidoglycan chain cannot grow. In other words, the elongation of the peptidoglycan and cross-linking of the chains is inhibited, thus leading to cell lysis. Hence, vancomycin is a bactericidal agent.

How does Vancomycin inhibit bacterial cell wall synthesis?

What Are the Adverse Effects of Vancomycin, and Bacitracin? (LO#2)

Use of antibiotics—although clearly a mainstay of medical therapy—is never without risk. Hypersensitivity, cross-reactivity, and end-organ damage are among the potential consequences of using powerful drugs such as cephalosporins or vancomycin. Let's discuss some specific negative effects we should always be on the lookout for.

Vancomycin

The two most important adverse effects that are important to note are the nephrotoxicity and ototoxicity. About 90% of the drug, vancomycin, is cleared renally (by glomerular filtration). There is also a risk of these adverse effects when vancomycin is paired with other nephrotoxic or ototoxic drugs, such as piperacillin-tazobactam. Thrombophlebitis and local infusion pain are far more common.

Nephrotoxicity

Vancomycin-induced nephrotoxicity is often the adverse effect that is most clinical relevant. The mechanism of vancomycin nephrotoxicity involves apoptosis induced by accumulation of drug in proximal tubular epithelial cells. Factors influencing risk of nephrotoxocity include dose, host-related factors (increased weight, pre-existing renal dysfunction, and critical illness), and concurrent administration of nephrotoxic agents. Nephrotoxic drugs such as aminoglycosides (aminoglycoside antibiotic), loop diuretics (furosemide) , amphotericin B (antifungal antibiotic), intravenous contrast dye, and vasopressors.)

Ototoxicity

Vancomycin can accumulate in the inner ear and cause ototoxicity. The ototoxicity risk increases when the drug is used concurrently with aminoglycosides and loop diuretics such as furosemide.

Red man syndrome

Red man syndrome is another side effect caused by diffuse flushing of the body. This is due to the effect of vancomycin on mast cells which causes them to release histamine. Antihistamines prior to vancomycin administration and slowing down the infusion rate are used to prevent occurrence of Red man syndrome. (Figure 6).

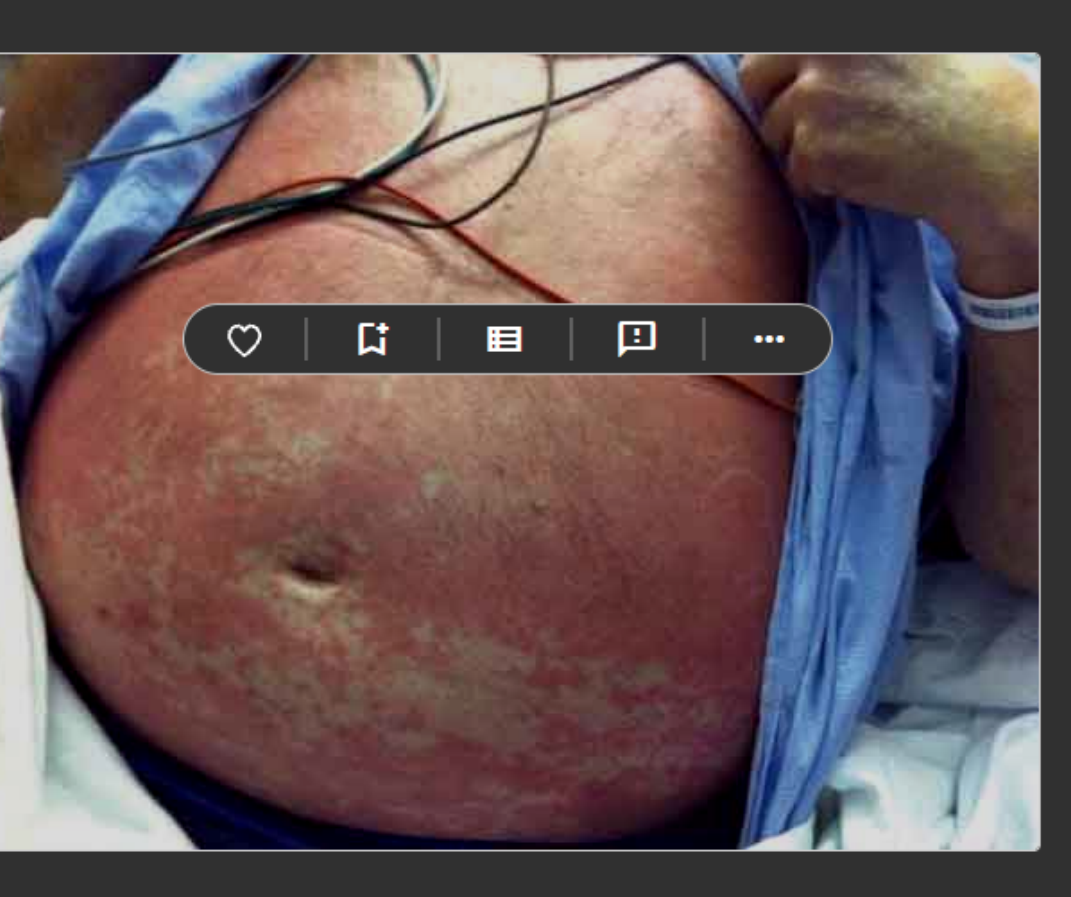


Figure 6

Bacitracin

Topical formulations can cause contact dermatitis. Opthalmic dosage can cause blurred vision.

CLINICAL CORRELATION

Vancomycin administration can cause common reactions, such as infusion-related flushing caused by the release of histamine. This can be prevented by prolonging the infusion period to 1–4 hours or pretreatment with an antihistamine such as diphenhydramine.

What measures can prevent Red man syndrome during vancomycin administration?

Other effects

Thrombophlebitis and local infusion pain are far more common.

CLINICAL CORRELATION

Most adverse reactions from IV Vancomycin use are relatively minor and reversible. Vancomycin-induced ototoxicity is rare, but nephrotoxicity is common. Vancomycin doses are now calculated using PK models that can utilize the AUC/MIC information to predict the dose based on individual and population-based parameters. Implementing these guidelines can reduce the occurrence of nephrotoxicity.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to DW, how do you explain your treatment plan?

You explain: AY has a skin infection with an extension to the subcutaneous tissue with cellulitis. The penetrating wound also increases the risk of the infection. The appearance of edema and tenderness also indicates that AY has a skin and soft tissue infection. The bugs that were tested were MRSA and group A β-hemolytic streptococci. The streptococcal infection can spread quickly, especially when the skin is broken. Treatment with IV Vancomycin will be a good option since it covers both bugs. Antibiotics that can cover MRSA are very limited. It is important to note that antibacterial stewardship is critical to prevent bacterial resistance.

Summary

- Vancomycin and bacitracin are peptidoglycan-inhibitor antibiotics that kill bacteria by breaking down their cell walls.
- The bacterial cell wall is made up of peptidoglycan, which includes a glycan (NAG or NAM) and a peptide bridge.
- Vancomycin and bacitracin inhibit peptidoglycan synthesis in different ways, affecting the construction of the bacterial cell wall.
- Vancomycin is used to treat serious gram-positive infections, including those caused by MRSA, while bacitracin is typically used for superficial skin infections.
- Adverse effects of vancomycin include nephrotoxicity, ototoxicity, and red man syndrome.

Review Questions

1. How does vancomycin inhibit peptidoglycan synthesis?

A. ○ Inhibiting transpeptidase
B. ○ Blocking transglycosylation
C. ○ Preventing NAM from joining bactoprenol
D. ● Binding to D-alanyl-D-alanine
E. ○ Inhibiting autolysins

Hide Explanation

The correct answer is binding to D-alanyl-D-alanine (D). Vancomycin inhibits peptidoglycan synthesis by binding to the D-alanyl-D-alanine precursor, blocking polymerization. Inhibiting transpeptidase (A) is the mechanism of beta-lactam antibiotics. Blocking transglycosylation (B) is not specific to vancomycin. Preventing NAM from joining bactoprenol (C) is the mechanism of bacitracin. Inhibiting autolysins (E) is not the mechanism of vancomycin.

2. Slow infusion of Vancomycin (IV) is administered in patients to avoid

A. ○ Diarrhea
B. ○ Headache
C. ○ Constipation
D. ● Red-man syndrome

Hide Explanation

The correct answer is Red-man syndrome (D). Red-man syndrome is caused by the release of histamine from mast cells causing flushing; hence the name. Diarrhea (A) and Headache (B) are non-specific reactions that can be caused by any IV antibiotic use. Constipation (C) usually does not occur with Vancomycin (IV) use.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain the bactericidal mechanism of action of vancomycin and bacitracin.~~
02 ☑ ~~Recognize the most common the adverse drug reactions associated with the use of vancomycin and bacitracin.~~

Objective Confidence: 2/2

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